SYMMET: Symmetry-Aware Machine Learning

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Abstract

The paper investigates symmetry-aware machine learning architectures and explores whether incorporating geometric structures into neural networks can improve their performance and sample-effiency. Three models are compared using a synthetic dataset of two concentric circles: a standard Multilayer Perceptron (MLP) using raw coordinates, a radial MLP using distance from the origin, and a radial logistic regression classifier. The models are evaluated based on their training performance, accuracy and ability to learn from limited data. The results show that the radial MLP generally achieves higher accuracy with fewer training samples than the standard MLP, while radial logistic regression performs less effectively. These findings suggest that incorporating geometric knowledge into input representations can improve learning efficiency, although model complexity remains important for achieving strong classification performance.

1 Introduction

1.1 Background and motivation

Machine learning is a field of artificial intelligence in which computers learn patterns from the provided data instead of following instructions written for every possible situation. Neural networks are commonly used in machine learning because they can learn complex relationships between input and output. They are used in image recognition, language processing, medical research and many other areas. However, traditional neural networks often need large amounts of labeled data. When only a small amount of data is available learning can become more difficult [2].

One possible way to improve learning is to give the neural network some information about the structure of the problem before training begins [1, 2]. For example, imagine a machine learning model that must classify points belonging to two circles. A normal neural network receives the coordinates of each point, represented as $((x, y))$ values. It must learn how these coordinates relate to the correct class. However, circles has an important geometric property: every point on a (near) perfect circle has the same distance from its center. This means that instead of using two coordinates we can describe the important information using a single value, the distance from the center, also called the radius. By providing this information directly we make the classification problem easier for the model.

This idea is related to an important concept in machine learning called inductive bias. An inductive bias is an assumption built into a learning system that helps it learn certain patterns more easily [1]. In this research we investigate whether adding knowledge about circular geometry can help machine learning models learn with fewer training examples.

1.2 Research Question

The main research question of this paper is: Does encoding geometric symmetry into a machine learning model improve its sample efficiency and classification performance? In this context, sample efficiency refers to how much training data a model needs to achieve a certain level of accuracy. A model with higher sample effiency can achieve similar performance with fewer training examples. To investigate this question, the study compares three models with different architectures and input representations.

The first model uses the original coordinates of each point (ModelA). The second uses the distance from the center of the circles (ModelB). The third uses the same radial information but applies a simpler classification method (ModelC). By comparing these models, the study aims to understand whether the representation of data can be as important as the complexity of the model itself [5].

1.3 Hypothesis

The main hypothesis is that a neural network receiving geometric information about the problem will learn more efficiently than a neural network receiving only raw coordinates. More specifically we expect a model using radial distance to require fewer training examples to learn the difference between the inner and the outer circles. Another hypothesis is that a simple classifier using the correct geometric representation may perform similarly or even better than a more complex neural network. The hypotheses are tested by training the models on datasets of different sizes and comparing their classification performance.

1.4 Objective of the study

The main objectives of this research are to one investigate how geometric symmetry affects machine learning. Two to compare a standard neural network with models that use radial information. Three, to measure how classification accuracy changes with training dataset sizes Four, to examine whether a simple model with useful geometric features can compete with a larger neural network. Five, to understand the benefits and limitations of including geometric assumptions in machine learning models.

2 Background and Related Work

2.1 Symmetry in Machine Learning

Symmetry is a mathematical property in which an object or a relationship remains unchanged under certain transformations [1, 2]. For example, a circle which looks the same when rotated around its center. A point on a circle can be moved to another position by rotating it but its distance from the center remains unchanged. Consider a point with coordinates: $p = (x, y)$. Its distance from the origin is calculated using:

$$r = \sqrt{x^2 + y^2} \tag{1}$$

If the point is rotated around the origin, the new coordinates will be different but its radius will remain the same. This property is called rotational invariance [2]. Rotational invariance means that a value or prediction does not change when the input is rotated [2]. For machine learning model rotational invariance can be written as: $f(R_\theta x) = f(x)$. Where R_θ represents a rotation by angle θ and f represents the models prediction function [2].

3 Methodology

3.1 Overview

The experiment compares three machine learning models on a binary classification problem. Each model must classify points belonging to one of two concentric circles [5]. The models as discussed before are:

- Model A: Baseline Multilayer Perceptron
- Model B: Radial Multilayer Perceptron
- Model C: Radial Logistic Regression

The main difference between the models is the information they receive and the complexity of their architecture. Model A receives the original coordinates of each point. Model B and Model C receive the radial distance calculated from those coordinates. The models are trained and evaluated using the same underlying dataset [5]. By comparing their performance, the experiment investigates whether a model benefits from receiving information about the geometry of the problem.

3.2 Dataset

The dataset is generated using the `make_circles` function from the Python library scikit-learn [4]. This function creates a synthetic dataset containing two concentric circles. Each data point has two coordinates $X = (x, y)$ and each point has a binary label $y \in (0, 1)$.

The label identifies which circle the point belongs to. The dataset contains 50,000 generated samples [5]. A noise value of 0.04 is used to make the classification problem less perfect and more similar to situations where data contains small variations. The circle factor is 0.8 which controls the size of the inner circle relative to the outer circle. Then the dataset is divided into training and testing portions. 80% of the data is used for training and 20% is reserved for testing to prevent overfitting. The training data is used to update the model parameters, while the testing data is used to evaluate performance on examples. The dataset is suitable for this investigation because its geometric structure is known.Below is a figure of what the raw dataset looks like.

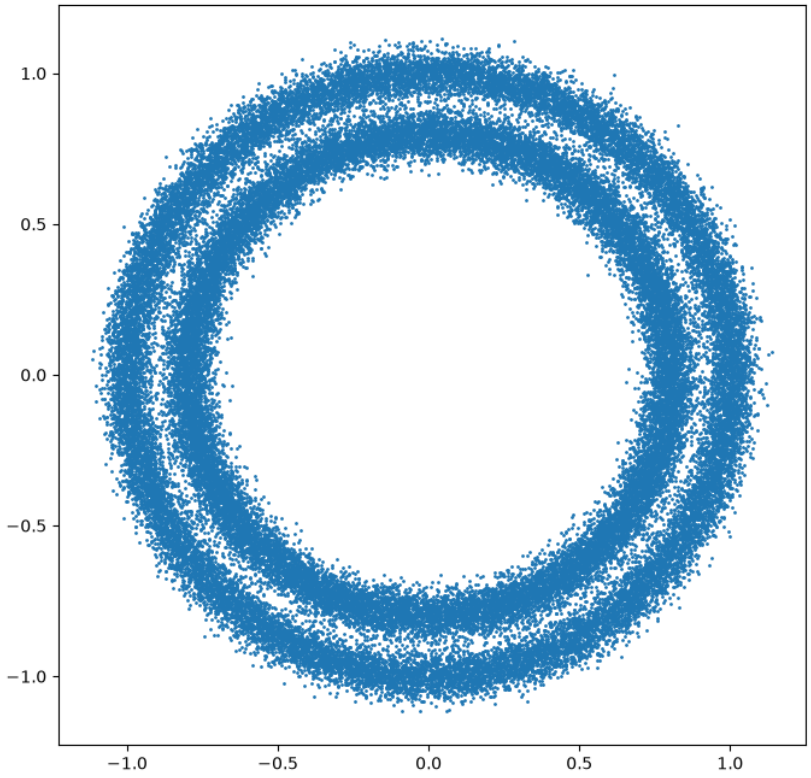


Figure 1: Synthetic dataset containing two concentric circles [5].

3.3 Model A: Baseline Multilayer Perceptron

The first model is a standard multilayer perceptron, also called an MLP. An MLP is a type of artificial neural network that uses layers of connected neurons to learn relationships between inputs and outputs. The baseline model receives two input values: $X_A = (x, y)$ these are the original coordinates of each point. The model contains two hidden layers, each with 16 neurons and Rectified Linear Unit (ReLU) is used as the activation function in the these layers. The output layer contains one neuron, which produces a value used for binary classification, this gives the model a total of 337 parameters [5]. The main purpose of this model is to provide a baseline for comparison. It does not receive any direct information about the radius of the circles. Instead it must learn the relationship between the coordinates and class labels through training. Although the network can learn nonlinear decision boundaries, it does not automatically produce a circular boundary.

3.4 Model B: Radial Multilayer Perceptron

The second model uses the same general MLP structure as the baseline model but instead of receiving the original coordinates, it receives the distance of each point from the origin. The input is calculated with Equation (1): $r = \sqrt{(x^2 + y^2)}$. Therefore, the model receives one input value compared to two on Model A. Just like the previous model, this one has 2 hidden layers with 16 neurons each and uses ReLU as the activation function. This model has a total of 321 trainable parameters [5]. The key difference between Model A and Model B is the input representation. Model A must learn how the coordinates relate to the circular classification boundary, whilst Model B receives information that already describes the most import geometric feature of the dataset. This means the model only needs to learn how radial distance relates to the class label. The experiment investigates whether this difference allows Model B to learn more efficiently.

3.5 Model C: Radial Logistic Regression

The third model is the logistic regression classifier that also receives radial distance as its input. Like Model B the input is $r = \sqrt{x^2 + y^2}$ however, Model C does not contain hidden neural network layers. Instead, it simply uses a single linear transformation: $z = wx + b$. Were w represents the weight and b represents the bias. The output can then be converted into a probability using the sigmoid function in:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{2}$$

The model contains only two trainable parameters [5]. This makes Model C much simpler than the other models. The purpose of including logistic regression is to investigate whether a complex neural network is necessary when the input already contains useful geometric information. Because the classification task involves separating points according to their distance from the center, a simple decision threshold on the radius can be sufficient. By comparing this model to the two MLPs, the study investigates the relationship between model complexity and input representation.

3.6 Model Comparison

The three models differ in their inputs and numbers of trainable parameters [5].

Table 1: Comparison of the three machine learning models

Model	Input	Architecture	Parameters
A	Coordinates (x, y)	MLP	337
B	Radius (r)	MLP	321
C	Radius (r)	Logistic Regression	2

Model A and B have similar numbers of parameters, making them useful for comparing the effects of the input representation. Model C is much smaller and is included to test whether the radial representation makes a simpler classification method sufficient. Together, the three models allow the experiment to compare both representation and model complexity.

3.7 Training Procedure

The models are implemented using PyTorch, an open-source machine learning framework [3]. Training is performed using the Adam optimizer. Adam is an optimization algorithm that updates a model's parameters in order to reduce prediction errors. The learning rate is set to $\eta = 0.001$. The training algorithm also makes use of mini-batch Adam optimization with a batch size of 128 [5].

Binary Cross-Entropy with Logits Loss is uses to measure the difference between predicted outputs and actual class labels, the loss function is suitable for binary classification problems. The model comparison experiment trains the models for 20 epochs [5]. The same general training procedure is used for the different models so their performance can be compared under similar conditions for the best results.

3.8 Evaluation Metrics

The experiment mainly focuses on classification accuracy and sample efficiency.

3.9 Classification Loss

The loss measures how closely a model's predictions match the correct labels. Lower loss values indicate that the model's predicted outputs better match the training or testing labels. Accuracy and loss are both useful because they provide different information about the model's performance.

3.10 Sample Efficiency

Sample efficiency measures how much training data a model needs to achieve a particular level of performance. For example, suppose two models achieve the same training accuracy. If the first model needs 1,000 training examples while the second model needs only 100, the second model is more sample-efficient at the accuracy level. This experiment evaluates the models using different training dataset sizes. This makes it possible to observe how quickly their performance improves as more data becomes available [5].

4 Methodology

4.1 Experimental Setup

The research includes two main experiments. The first experiment compares the training behavior and classification performance of the three models. The second experiment investigates sample efficiency by training the models using different amounts of data. The goal is to observe whether incorporating geometric information changes how much data the model needs to lean the classification problem.

4.2 Experiment 1: Model Performance Comparison

In the first experiment, all three models are trained on the concentric-circle dataset. The training process runs for 20 epochs. During training, the classification accuracy and loss of each model are recorded. These values are used to generate graphs showing how performance changes over the training process [5].

The Experiment is designed to compare the learning behavior of the baseline MLP, radial MLP, and radial logistic regression model. An important part of this comparison is determining whether the radial models reach good classification performance in fewer epochs. Another import par is comparing the final accuracy of each model.

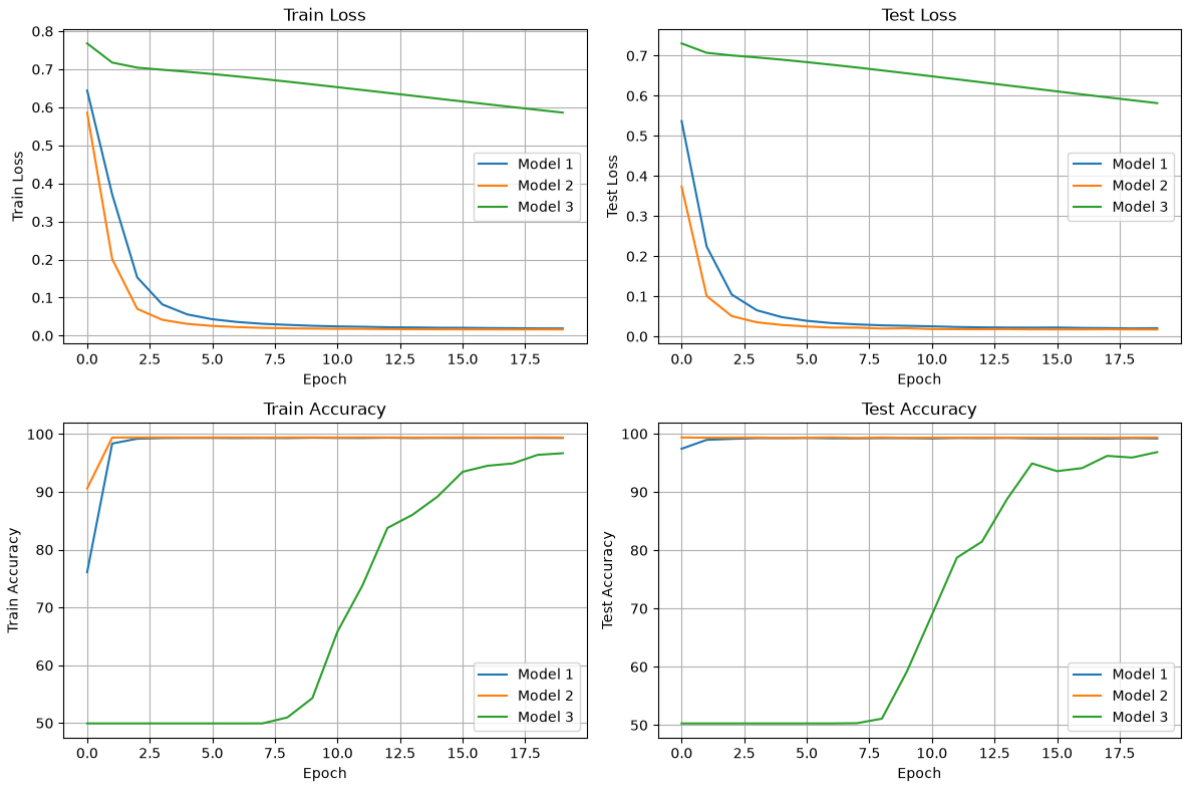


Figure 2: Training and testing accuracy of Models A, B and C over 20 epochs [5]. (More detailed images are on the GitHub repository of this research).

The graphs show that both Models A and B reduce their loss rapidly, indicating that they learn the underlying patterns in the data early in the training. Their loss curves then gradually flatten out as the models converge, which is expected behavior. Model B (Orange function) decreases the loss function a little faster than Model A which lines up with the Hypothesis although this does not mean that Model A is automatically more sample-efficient. There can be multiple causes to a better loss curve, for example a faster optimization. On other hand Model C learns at a much slower rate, which is understandable given that it only has two trainable parameters [5].

What is particularly interesting about Model C is that its loss remains relatively high despite achieving a high accuracy. This can be explained by the Binar Cross-Entropy with logits function, which unlike accuracy does not simply measure whether a prediction is correct but also takes the model's confidence into account [3]. While accuracy only considers whether the predicted class matches the true label, the loss function penalizes predictions based on their predicted probabilities. The difference between Model C's loss and accuracy therefore suggests that the model correctly classifies most samples but does so with relatively low confidence. As a result, Model C can achieve high accuracy while still maintaining a comparatively high loss [5].

4.3 Decision Boundary Analysis

A decision boundary separates the regions that a classification model assigns to different classes. For this experiment the ideal boundary is approximately circular because the data consists of concentric circles. The base line MLP learns its boundaries from the original coordinates. The radial MLP and radial logistic regression model use distance from the origin, which gives them circular boundaries by construction [2, 5].

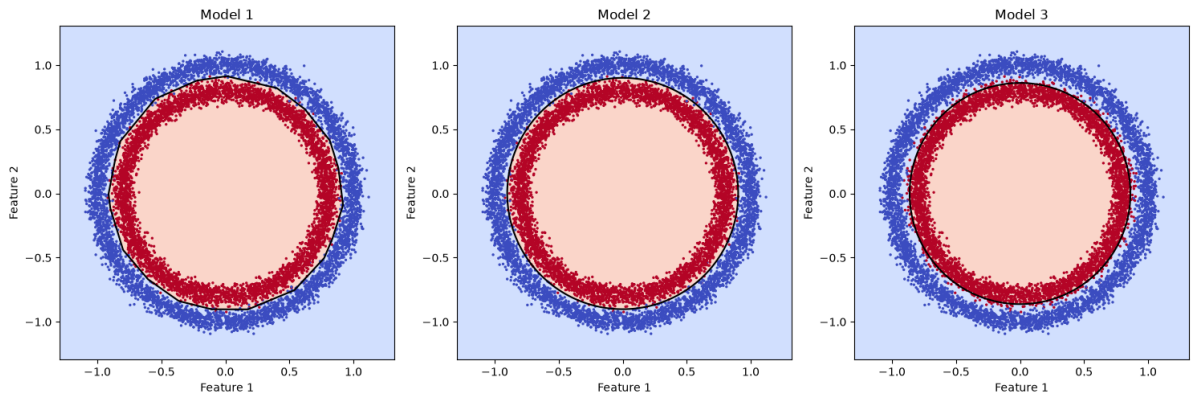


Figure 3: Decision boundaries produced by the three classification models [5]. (More detailed images are on the GitHub repository of this research paper).

The radial models produce well defined, nearly perfect circular decision boundaries. As illustrated in Figure 3, Model B and C exhibit considerably clearer and more distinct boundaries than Model A. Furthermore, Model B produces a larger decision boundary than Model C, encompassing a greater proportion of the inner data points. This difference is the cause of variations in the model's accuracy, as Model B demonstrates higher accuracy than Model C.

4.4 Experiment 2: Sample Efficiency

The second experiment investigates how the models perform when the available training data is limited. Each model is trained using different numbers of training examples. The dataset sizes are: 20, 50, 100, 250, 500, 1,000, 2,500 and 10,000 [5]. For each dataset size, the model is used to create a simple efficiency graph.

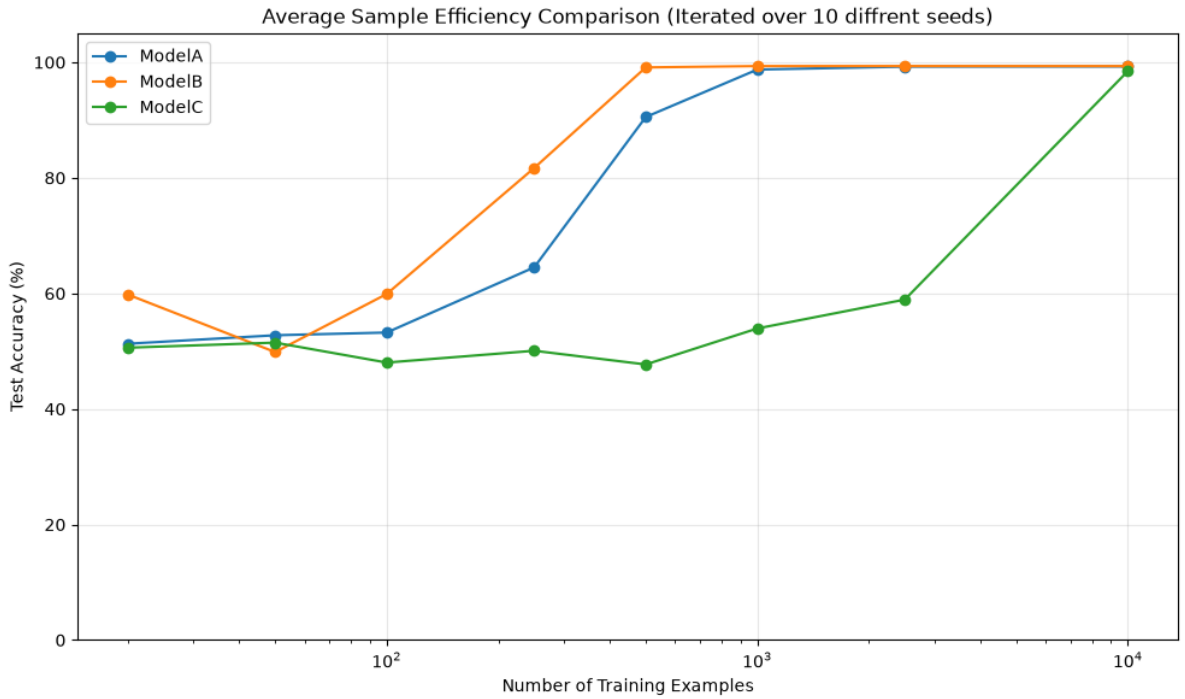


Figure 4: Test accuracy of the three models for different training datasets [5]. (More detailed images are on the GitHub repository of this research paper).

Figure 4 shows that both Model A and Model B achieve reasonable accuracy. However the main difference is how quickly they reach this accuracy. Model B reaches higher accuracy faster than Model A and generally performs better. Before (10^2) on the y-axis, all the three models show similar performance. This is expected, as the models have not yet received enough data to learn effective patterns. After (10^2) the differences become more noticeable. Both Model A and Model B begin to improve model Mode B has a little faster improvement and a higher Accuracy, making it the more sample-efficient model of the two. In contrast, Model C learns at a much slower rate and does not reach the same accuracy as the other two models. This may be explained by the fact that Model C has only two learnable parameters, which limits its ability to learn complex patterns in the data [5].

4.5 Summary of Experimental Evidence

The two experiments investigate different aspects of learning. The model comparison experiment examines how the architectures behave during training and how they represent the classification boundary. The sample-efficiency experiment provides a way to study the relationship between geometric information, model complexity and learning performance. Together these experiments show that the main hypothesis is partially supported. The results suggest that a model using radial distance as input is more sample-efficient and learns faster than a model using only raw coordinates. This shows that providing geometric information can help a neural network learn more efficiently. However, the second hypothesis is not supported by the results. This suggests that while using the

correct geometric representation can improve learning efficiency, model complexity is still important for achieving high accuracy [5].

5 Discussion

5.1 The importance of Input Representation

One of the main ideas investigated in this research is that the way data is represented can strongly affect how a machine learning model learns. A neural network receiving raw coordinates must learn how the two input values relate to the correct class. For a circular classification problem, the relevant relationship involves the distance from the center.

For other more complex problems where large amounts of labeled data may not be readily available, similar approaches to data representation can help models learn more efficiently. By incorporating relevant relationships or structures into the input features, models may require fewer training samples to achieve good performance, making the learning process more sample-efficient.

5.2 Limitations of the study

Although the experimental setup provides a useful comparison, it has several limitations. First the dataset is synthetic. Synthetic datasets are fully used for controlled experiments but do not contain all the complexity found in real-world data. Second, the study focuses only on two concentric circles, this means that geometric relationship is simple and know in advance. More complex classifications problems may not be described using a single radial feature. Third the radial models assume that the center of the circles is at the origin, if for example the circles were moved to a different location, the distance from the origin may no longer be the best feature for classification. Finally the models are not perfectly matched in architecture or parameter count.

5.3 Implications for Machine Learning

The broader significance of this research lies in the relationship between prior knowledge and machine learning. One of the fundamental principles of machine learning is that it is particularly useful for problems that are difficult to solve using explicitly defined rules. For example, recognizing handwritten digit, using traditional programing would require an extremely large number of rules and conditions to account for variations in handwriting, making such an approach impractical. However, this does not mean that traditional problem-solving methods are machine learning must be treated as entirely separate approaches. Instead of relying on a neural network to learn every aspect of a problem from scratch, existing mathematical knowledge and known relationships can be used to simplify parts of the task before training begins. This research explores this idea

by demonstrating how incorporating geometric knowledge into the input representations can reduce the amount of information a model needs to lean, potentially improving its learning efficiency.

6 Conclusion

This paper investigates the role of geometric symmetry in machine learning through a binary classification problem involving concentric circles. Three models are compared: a standard MLP using raw coordinates, a radial MLP using distance from the origin and a radial logistic regression model. The results show that incorporating geometric information into the input representation can improve learning efficiency and classifications performance. The radial MLP generally achieved higher accuracy with fewer training samples than the standard MLP, supporting the main hypothesis. However the radial logistic regression model did not achieve the same level of performance, suggesting that while an appropriate input representation is important, model architecture and complexity still play an important role.

The broader conclusion of this research is that machine learning does not always need to approach a problem entirely from scratch. When certain mathematical relationships or properties of a problem are already known, incorporating this knowledge into the learning process can reduce the amount of information a model must discover through training. Although this experiment focuses on a relatively simple geometric classification problem, the underlying principle may also be relevant to more complex applications where labeled data is limited or expensive to obtain.

References

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